

TRUCKLY

Statistical Analysis of Freight Truck Journey Optimisation

A simulation–optimisation study of empty return journeys on the Chennai · Bengaluru · Coimbatore freight corridor

1 · PROBLEM STATEMENT

A truck delivers Chennai → Bengaluru. With no return load it drives back EMPTY, or idles waiting for one. Both waste fuel, capacity, driver hours and money, and add avoidable vehicle-kilometres and CO₂.

Under the baseline policy modelled here 100% of return kilometres are empty: a mean of 150 km per truck and 4,495 km per 30-truck operating day.

2 · OBJECTIVE

RQ — Can intelligent freight matching and route optimisation reduce empty truck kilometres while keeping deliveries feasible and improving fleet efficiency?

Formally a SELECTIVE Pickup-and-Delivery Problem with Time Windows: choose which loads to accept, assign them to trucks and route each truck — minimising cost subject to capacity, time windows, detour tolerance and driver-hours limits.

3 · DATASET

△ THE FREIGHT RECORDS ARE SIMULATED. They are NOT observations of real Indian freight.

OBS 40 node coordinates, real (OpenStreetMap)
SIM 18,000 shipments · 3,000 truck-journeys
100 simulated operating days × 30 trucks
DRV distances · fuel · cost · utilisation · CO₂
ASM circuitry · speeds · fuel curve · tariffs · driver rules
Seeded and reproducible: same seed ⇒ identical data.

4 · DESCRIPTIVE STATISTICS

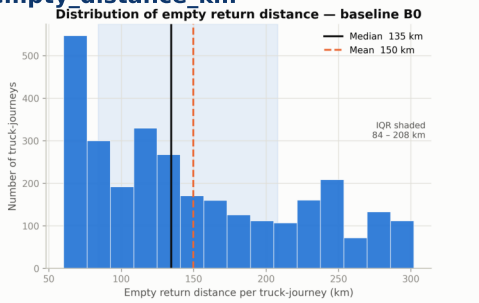
baseline B0 (empty return) · n = 3,000 truck-journeys · scenario S0_base

Variable	Unit	Mean	SD	Min	P10	Q1	Median	Q3	P90	Max	IQR
total distance km	km	149.8	71.5	60.2	65.2	84.2	134.6	208.1	255.2	302.0	123.9
empty distance km	km	149.8	71.5	60.2	65.2	84.2	134.6	208.1	255.2	302.0	123.9
fuel litres	L	38.2	20.7	8.6	15.7	21.2	33.5	51.6	69.0	116.4	30.4
fuel cost inr	INR	3,434.5	1,860.4	775.4	1,417.3	1,907.6	3,010.7	4,644.5	6,210.4	10,479.1	2,736.9
driving time hr	hours	2.9	1.2	1.3	1.4	1.8	2.8	4.0	4.6	5.5	2.3
duty time hr	hours	2.9	1.2	1.3	1.4	1.8	2.8	4.0	4.6	5.5	2.3
total cost inr	INR	6,556.9	2,105.7	3,555.0	4,218.6	4,784.9	6,094.5	7,981.1	9,696.2	14,176.1	3,196.2
co2 kg	kg	102.3	55.4	23.1	42.2	56.8	89.7	138.3	184.9	312.0	81.5

5 · STATISTICAL VISUALISATIONS

five required panels + one supplementary · the observation box under each panel is left blank for handwriting

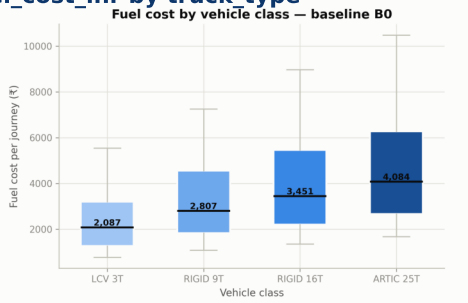
G1 · HISTOGRAM — empty distance km



STATISTICAL FINDING: right-skewed (skew 0.50); mean 150 km > median 135 km; IQR 124 km; top decile carries 19% of all empty kilometres

HANDWRITTEN OBSERVATION

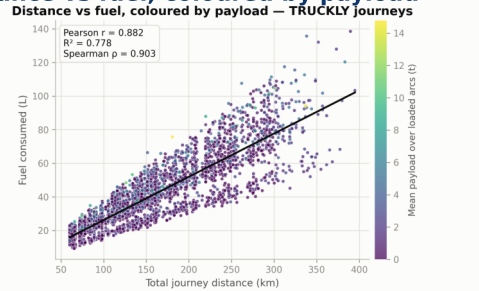
G2 · BOX PLOT — fuel_cost_inr by truck type



STATISTICAL FINDING: within-class IQR (~₹2,890) EXCEEDS the between-class median range (₹1,997); 0 points beyond the 1.5×IQR fence

HANDWRITTEN OBSERVATION

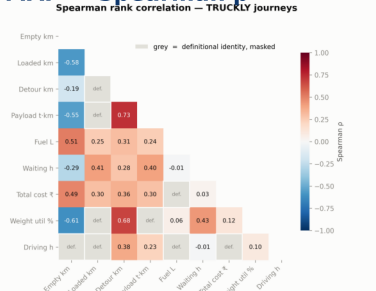
G3 · SCATTER — distance vs fuel, coloured by payload



STATISTICAL FINDING: Pearson $r = 0.882$, $R^2 = 0.778$, Spearman $\rho = 0.903$; relationship is PARTLY STRUCTURAL by construction — see caption

HANDWRITTEN OBSERVATION

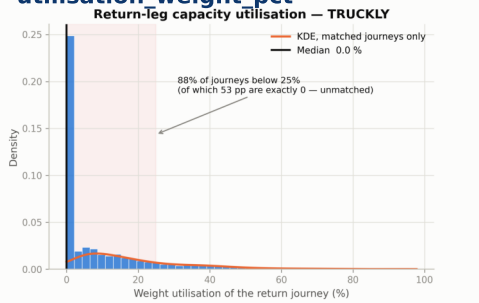
G4 · CORRELATION HEATMAP — Spearman ρ



STATISTICAL FINDING: strongest non-definitional pairs: Payload t-km↔Detour km $\rho = +0.73$, Weight util %↔Detour km $\rho = +0.68$; definitional identities are masked, not reported as findings

HANDWRITTEN OBSERVATION

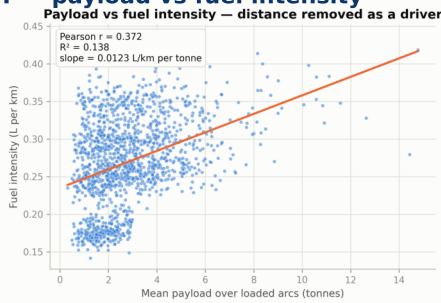
G5 · DISTRIBUTION — utilisation weight pct



STATISTICAL FINDING: BIMODAL: 53% exactly 0% (unmatched), median of matched legs 14.2%; 88% of legs below the 25% under-utilisation threshold

HANDWRITTEN OBSERVATION

G3b · SUPPLEMENTARY — payload vs fuel intensity



STATISTICAL FINDING: dividing fuel by distance removes the trivial driver: r falls $0.88 \rightarrow 0.37$, slope 0.0123 L/km per tonne carried

HANDWRITTEN OBSERVATION

6 · KEY FINDINGS — baseline vs TRUCKLY

R = 100 paired days · common random numbers

KPI	B0 empty	B2 greedy	TRUCKLY	vs B0	vs B2
Empty km	4,495.0	3,418.0	3,383.0	+24.7 %	+1.0 %
Total km	4,495.0	4,781.0	4,801.0	-6.8 %	-0.4 %
Fuel	1,144.8	1,243.4	1,249.2	-9.1 %	-0.5 %
Net cost (net)	196,706.0	163,620.0	160,173.0	+18.6 %	+2.1 %
Weight utilisation	0.0	6.3	6.8	n/a	+0.47 pp
Backhaul fill rate	0.0	47.4	47.3	n/a	-0.10 pp
Shipments served	0.0	14.2	15.9	n/a	+11.7 %
On-time rate	100.0	100.0	100.0	+0.00 pp	+0.00 pp

7 · CONCLUSION

Optimised matching cut empty running by 24.7% and net cost by 18.6% versus the empty-return baseline, and beat a NAIVE GREEDY matcher by 1.01% on empty km (Wilcoxon $p < 0.001$, lower in 69/100 days) and 2.1% on net cost.

The 60% aspiration was NOT reached on any metric. The benefit is not a constant — it rises from 10% at 2 loads/truck to 44% at 15 loads/truck.

Fuel and gross cost ROSE (-9.1 %, -7.4 %): carrying freight burns more than running empty. The gain is in NET cost and in utilisation.

8 · DATA PROVENANCE AND LIMITATIONS

OBS observed / external — 40 node coordinates from OpenStreetMap / public gazetteer.
SIM simulated — ALL shipments, trucks, weights, time windows and journeys.
DRV derived — distances, fuel, cost, utilisation, CO₂ (from OBS + SIM + ASM).
ASM assumed — road circuitry 1.18–1.35×, truck speeds, fuel curve, diesel ₹90/L, tolls, wages, driver-hour rules, freight tariff, CO₂ factor 2.68 kg/L.
Every value is declared in config.yaml.

LIMITATIONS

- No live routing engine was reachable from the build environment: road distance is great-circle × a declared circuitry factor, NOT traced road geometry.
- No real carrier freight records, no live traffic, no cancellations, no competing carriers.
- Single-leg return model; multi-day journey chains are out of scope.
- CO₂ is ESTIMATED from an assumed emission factor — it is not a measurement.
- Results are conditional on the generator: they describe the SIMULATED environment.